

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 3: Hardware

Subtopic 3.2: Logic Gates and Logic Circuits

Concept 3.2.4: Constructing Logic Expressions

Total Questions: 12

Mark Schemes begin on Page 9

Question 1 | Source: 9618_w25_qp_11 (Q3.b)

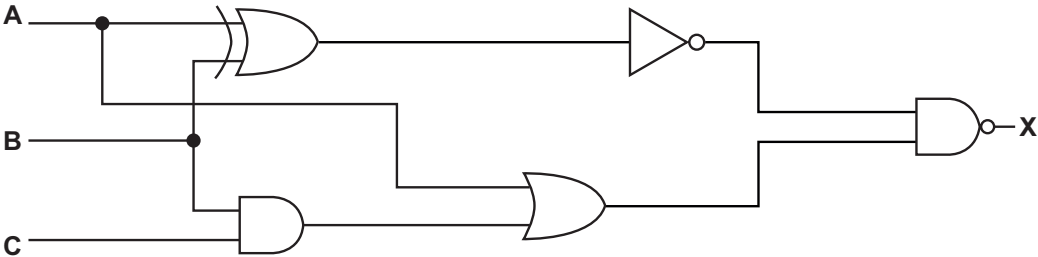
(b) Write the logic expression for the following truth table.

R	S	T	Q
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

Q = [2]

Question 2 | Source: 9618_w25_qp_12 (Q9.a)

9 (a) Consider the following logic circuit:

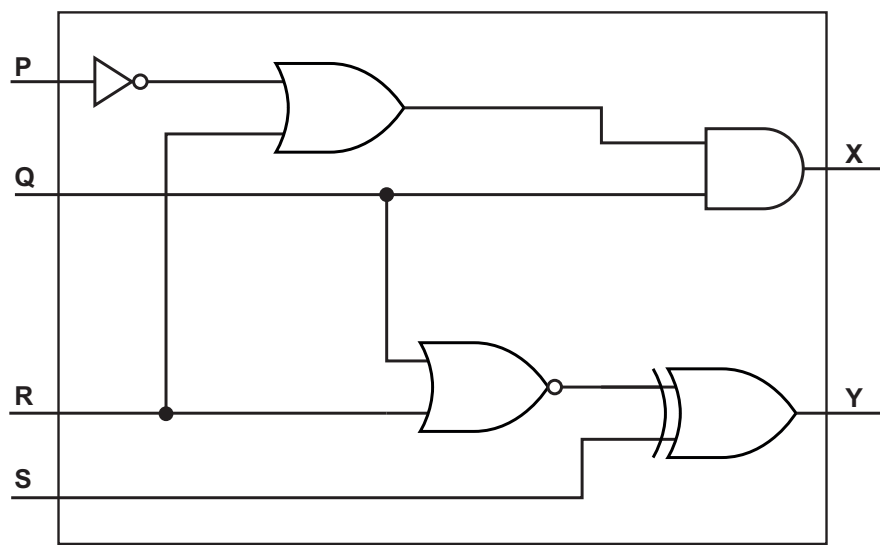


Write the logic expression for the logic circuit. Do **not** simplify the expression.

X = [2]

Question 3 | Source: 9618_s25_qp_11 (Q1.a)

1 (a) Write the logic expressions for the following logic circuit.



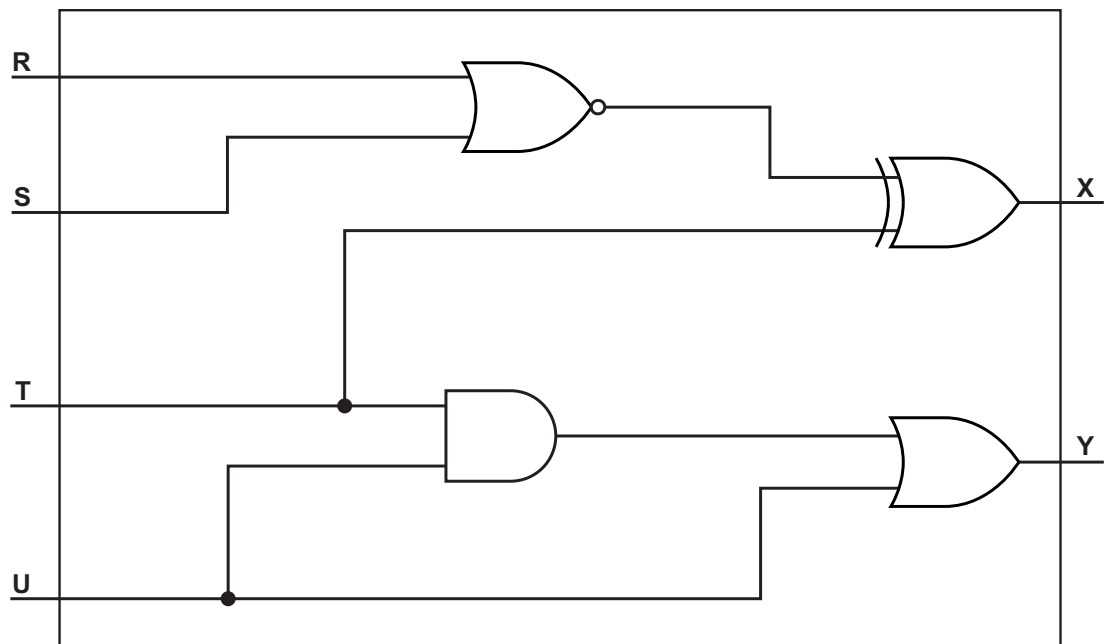
X =

Y =

[2]

Question 4 | Source: 9618_s25_qp_12 (Q4.a)

4 (a) Write the logic expressions for the following logic circuit.



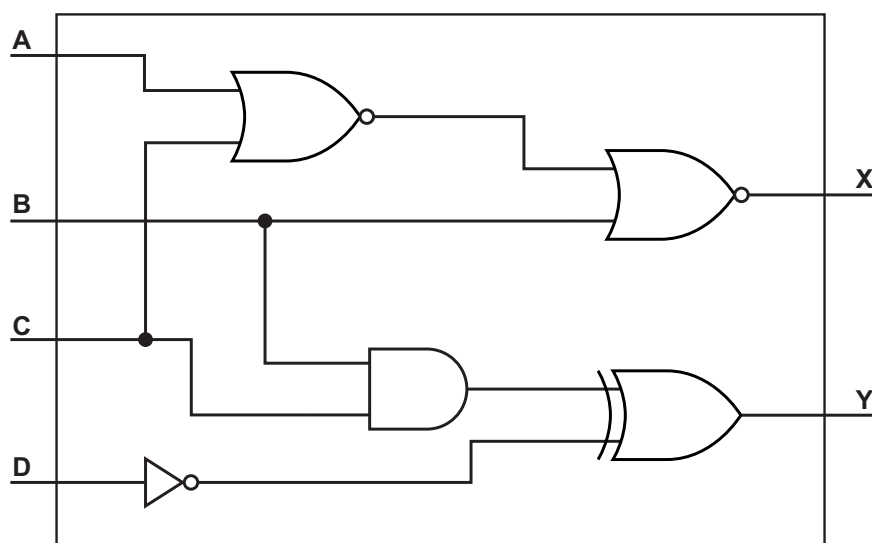
X =

Y =

[2]

Question 5 | Source: 9618_s25_qp_13 (Q4.a)

- 4 (a) Write the logic expressions for the following logic circuit.



X =

Y =

[2]

Question 6 | Source: 9618_w24_qp_11 (Q5.b)

- 5 A security system has both a floodlight (very bright light) and an audio alarm.

The data from multiple sensors is analysed and used to:

- turn on the floodlight
- sound the audio alarm.

Sensors can be used to detect:

- if doors are open
- the external daylight level
- if people are detected within a set distance.

(b) The floodlight (**X**) and audio alarm (**Y**) operate according to the following criteria:

Parameter	Description of parameter	Binary value	Condition
A	external daylight level	1	Low
		0	High
B	front door	1	Open
		0	Closed
C	person is within 2 m	1	Detected
		0	Not detected
D	back door	1	Open
		0	Closed
E	security system	1	Switched on
		0	Switched off

The floodlight turns on (**X = 1**) if:

- the security system is switched on
and
- the external daylight level is low
and
- a person is detected within 2 m.

The audio alarm turns on (**Y = 1**) if:

- the security system is switched on
and
- one **or** more doors are open, **or** a person is detected within 2 m.

Write logic expressions for the security system.

X =

.....

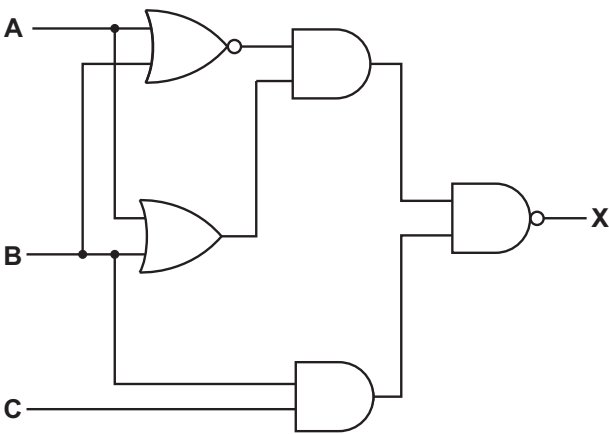
Y =

.....

[2]

Question 7 | Source: 9618_w24_qp_13 (Q1.b)

(b) Write the logic expression for the following logic circuit.



X =
..... [2]

Question 8 | Source: 9618_s24_qp_11 (Q1.a)

1 (a) Tick (✓) **one** box to identify the correct logic statement for this truth table.

A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

- ☐ NOT (A AND B AND C)
- ☐ (A XOR B) NOR C
- ☐ (A OR B OR C) NOR C
- ☐ NOT A AND NOT B AND NOT C

[1]

Question 9 | Source: 9618_s24_qp_13 (Q6)

- 6 Each of the following truth tables has three inputs (**A**, **B** and **C**) and one output (**X**).

Draw **one** line to match each truth table with its logic expression.

Truth table

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

Logic expression

NOT (**A** XOR **B**) AND **C**

(**A** OR **C**) AND NOT **B**

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

A NAND **B** NAND **C**

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

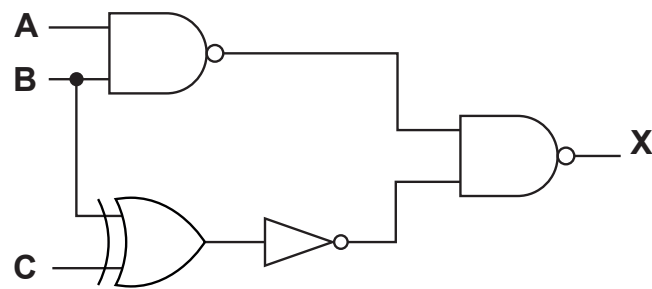
(**A** NAND **B**) OR **C**

NOT (**A** AND **B** AND **C**)

[3]

Question 10 | Source: 9618_w23_qp_12 (Q4.a)

4 (a) Write the Boolean expression that corresponds to the following logic circuit.



Question 11 | Source: 9618_s23_qp_13 (Q1.a)

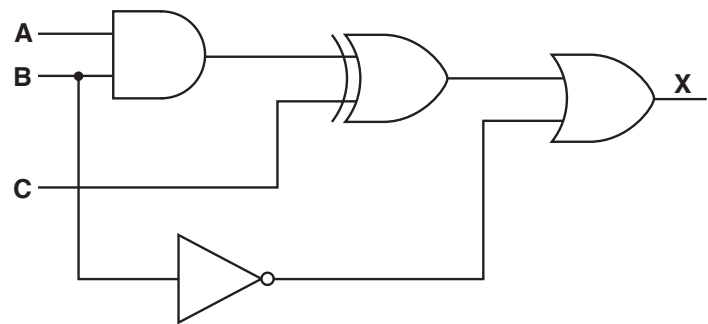
1 (a) Write the logic expression for this truth table:

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

..... [1]

Question 12 | Source: 9618_w21_qp_11 (Q3.a)

3 A logic circuit is shown:



(a) Write the logic expression for the logic circuit.

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_11 (Q3.b)

3(b)	1 mark per bullet point, max 2 marks - One correct term: (R AND S AND NOT T) // (NOT R AND NOT S AND T) - Second correct term plus the OR in correct place Answer: Q = (R AND S AND NOT T) OR (NOT R AND NOT S AND T)	2
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Mark Scheme for Question 2 | Source: 9618_w25_ms_12 (Q9.a)

9(a)	1 mark per bullet point, max 2 marks - NOT (A XOR B) - NAND ((B AND C) OR A) X = (NOT (A XOR B)) NAND ((B AND C) OR A) // X = ((B AND C) OR A) NAND (NOT (A XOR B))	2
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Mark Scheme for Question 3 | Source: 9618_s25_ms_11 (Q1.a)

1(a)	1 mark each: X = Q AND (NOT P OR R) Y = S XOR (Q NOR R)	2
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Mark Scheme for Question 4 | Source: 9618_s25_ms_12 (Q4.a)

4(a)	1 mark each: - X = (R NOR S) XOR T - Y = (T AND U) OR U	2
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Mark Scheme for Question 5 | Source: 9618_s25_ms_13 (Q4.a)

4(a)	1 mark for each correct statement X = B NOR (A NOR C) Y = (B AND C) XOR NOT D	2
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Mark Scheme for Question 6 | Source: 9618_w24_ms_11 (Q5.b)

5(b)	1 mark for each correct expression $X = E \text{ AND } A \text{ AND } C$ $Y = E \text{ AND } (B \text{ OR } C \text{ OR } D)$	2
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Mark Scheme for Question 7 | Source: 9618_w24_ms_13 (Q1.b)

1(b)	1 mark each: • $((A \text{ NOR } B) \text{ AND } (A \text{ OR } B))$ • $\text{NAND } (B \text{ AND } C)$ $X = ((A \text{ NOR } B) \text{ AND } (A \text{ OR } B)) \text{ NAND } (B \text{ AND } C)$	2
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Mark Scheme for Question 8 | Source: 9618_s24_ms_11 (Q1.a)

1(a)	1 mark for: $(A \text{ XOR } B) \text{ NOR } C$	1
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Mark Scheme for Question 9 | Source: 9618_s24_ms_13 (Q6)

6

1 mark for each correct line:

3

Truth table				Logic expression
A	B	C	X	
0	0	0	0	NOT (A XOR B) AND C
0	0	1	1	
0	1	0	0	
0	1	1	0	
1	0	0	1	
1	0	1	1	(A OR C) AND NOT B
1	1	0	0	
1	1	1	0	

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

				(A NAND B) OR C
				NOT (A AND B AND C)

Mark Scheme for Question 10 | Source: 9618_w23_ms_12 (Q4.a)

4(a)	1 mark for each bullet point - A NAND B - NOT(B XOR C) - Final NAND X = A NAND B NAND (NOT(B XOR C))	3
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Mark Scheme for Question 11 | Source: 9618_s23_ms_13 (Q1.a)

1(a)	NOT B	1
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Mark Scheme for Question 12 | Source: 9618_w21_ms_11 (Q3.a)

3(a)	1 mark per bullet point • A AND B ... • ... XOR C ... • ... OR NOT B ((A AND B) XOR C) OR NOT B	3
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